

LINEAR EQ:

LINEAR ALGEBRA. (MT-1004)

based upon.
Linear algebra $\hat{=}$ Linear eq.s.
"An eq. is said to be linear if power of each variable is 1 and no product of two variables appear."

FOR EX:-

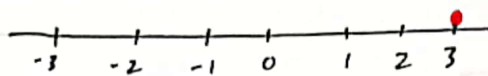
$$3x + 4y = 5$$

variable transcendent.

$$\begin{aligned} xy &= 1 \\ y &= \frac{1}{x} \\ y &= x^{-1} \end{aligned}$$

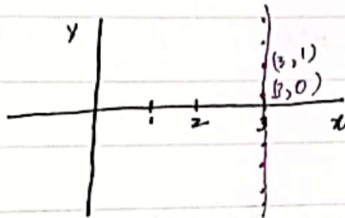
GEOMETRIC REPRESENTATION OF $x=3$:

In 1D:



so $x=3$ is a point in 1D

In 2D:



so $x=3$ is a line in 2D. the eq is

$$x + 0y = 3$$

vary/variable but not zero because

$$x + 0(1) = 3$$

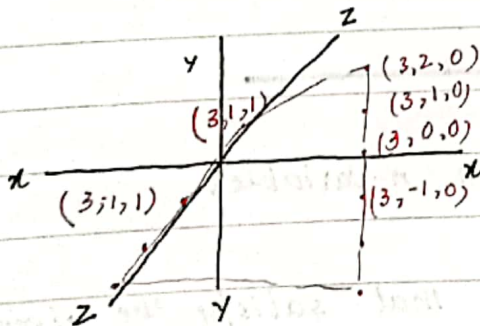
$$x + 0(4) = 3$$

$$x + 0(2) = 3$$

it is zero after multiplication
varying but not zero

$x=3 \rightarrow$ point
 $x+0y=3 \rightarrow$ line
 $x+0y+0z=3 \rightarrow$ plane

In 3D:



$$x + 0y + 0z = 3$$

$$x + 0(1) + 0(1) = 3 \Rightarrow \text{varying}$$

$\Rightarrow$ we can write many variables due to restrictions etc. numbers as.

$$x + y + z + k + f \dots = 1 \quad \text{or} \quad \begin{array}{cccc} x_1 & + & 2x_2 & + & 3x_3 & + & 4x_4 & \dots \end{array}$$

$x \quad \quad y \quad \quad z \quad \quad \dots$

$\Rightarrow$ if any coefficient is zero then we obtain vertical line as
 $x_1 + 0x_2 = 1$ or $0x_1 + 3x_2 = 1$ otherwise any line shape of direction is obtained according to intercepts value.

SOLUTION:-

These values of unknowns that satisfy given eq.

This two eq. &
three unknowns.

$$\Leftrightarrow \begin{cases} x_1 - 2x_2 = 5 \\ x_1 + 3x_3 = 0 \end{cases}$$



SYSTEM OF LINEAR EQ:-

"A collection of one or more than one linear eq. is called"

R EX: $x_1 - 2x_2 = 1$ $\Rightarrow$ This is called system of linear eq. which has one eq. and two unknowns.

• $\begin{cases} x_1 - 2x_2 = 5 \\ -x_1 + x_2 = 3 \end{cases} \Rightarrow$ This is called which has two eq. and two unknowns.

• $\begin{cases} x_1 - x_2 + x_3 = 5 \\ -x_1 + x_2 + 3x_3 = 0 \end{cases} \Rightarrow$ This which has two eq. and three unknowns.

$\Rightarrow$ $\begin{cases} x_1 + x_2 + 0x_3 = 3 \\ x_1 + x_2 + 0x_3 = 1 \end{cases} \Rightarrow$ Geometry is plane because we are dealing it with three unknowns so we tell that in first eq. there is a hidden third unknown as zero.

$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = b$
This is called linear eq. in/with n -variables

SOLUTION:-

Those values of unknowns that satisfy the given system.

Ex Is $(3, 4, -2)$ a soln. of following system.

$5x_1 - x_2 + 2x_3 = 7$	$\stackrel{\text{soln}}{=} 7 = 7$	So, $(3, 4, -2)$ is a soln. of given system.
$-2x_1 + 6x_2 + 9x_3 = 0$	$0 = 0$	
$-7x_1 + 5x_2 + 3x_3 = -7$	$-7 = -7$	

This system has three planes.

• soln. is just the point of intersection.

A GIVEN SYSTEM OF LINEAR EQ has either:

- Unique soln.
- Infinitely many soln.
- No soln.

• when we multiply an eq. or line with -ve sign than no direction or orientation of line is changed but if we

Ex. 1

$$x_1 - 2x_2 = -1$$

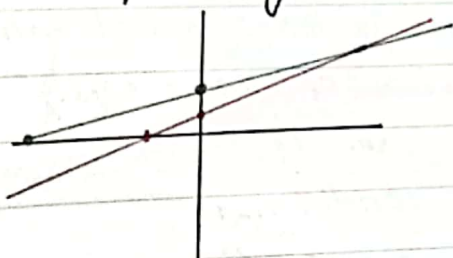
$$-x_1 + 3x_2 = 3$$

soln

Mathematically, $x_2 = 2 \Rightarrow x_1 - 4 = -1 \Rightarrow x_1 = 3$

so, $(3, 2)$ is a soln.

Graphically,



for 1st eq. $(0, \frac{1}{2}), (-1, 0)$
for 2nd eq. $(0, 1), (-3, 0)$
multiply vector with -ve sign than direction changes.

Ex. 2

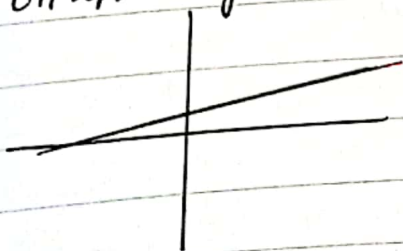
$$x - 2x_2 = -1$$

$$-x_1 + 2x_2 = 1$$

soln

mathematically, by adding $0=0$ (True statement)

Graphically,



for 1st eq. $(0, \frac{1}{2}), (0, -1)$

for 2nd eq. $(0, \frac{1}{2}), (0, -1)$

As both lines overlap so, given system has infinitely many soln.

$\Rightarrow$ If we multiply the given eq. with -ve sign than there is no effect on its original orientation.

Ex. 3

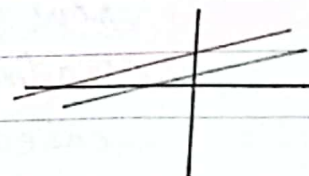
$$x_1 - 2x_2 = -1$$

$$-x_1 + 2x_2 = 3$$

soln

By adding $0=2$ (wrong statement).

Graphically it is parallel so they have no soln.



MATRIX NOTATION:-

consider a system of linear eq.

$$5x_1 - 2x_2 + x_3 = 2$$

$$-x_1 + x_2 + 5x_3 = 1$$

$$2x_1 - x_2 = 7$$

The augmented matrix of this system is.

$$\left[\begin{array}{ccc|c} 5 & -2 & 1 & 2 \\ -1 & 1 & 5 & 1 \\ 2 & -1 & 0 & 7 \end{array} \right] \Rightarrow \text{if this line is removed then it is co-efficient matrix.}$$

these are co-efficients.

ECHELON FORM:-

matrix inverse method $= x = A^{-1}b$

(To Find the soln. of system of linear eqs.)

"A given matrix is an Echelon form if it satisfy the following three conditions:

1. All non-zero rows are above any row of all zeros.
2. Each leading entry of a row lies towards the right of the leading entry of the row above it.
3. All entries below the leading entry must be zero."

Ex

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

leading entry / pivot point. in first row.

in second row

non zero row

zero row

LEADING ENTRY:-

"The first non zero element of a row is and the position of this leading entry in the matrix is called pivot position."

Ex.1

CONVERSION OF MATRIX INTO ECHELON FORM:-

$$x_1 - 2x_2 + x_3 = 0$$

$$2x_2 - 8x_3 = 8$$

$$-4x_1 + 5x_2 + 9x_3 = 9$$

The augmented matrix is.

$$= \begin{bmatrix} \boxed{1} & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & 9 \end{bmatrix} \Rightarrow \begin{bmatrix} \boxed{1} & -2 & 1 & 0 \\ 0 & \boxed{2} & -8 & 8 \\ \boxed{0} & -3 & 19 & -9 \end{bmatrix}; \begin{array}{l} R_3 \pm \boxed{} R_1 \\ R_3 \pm 4 R_1 \\ -4 + 4(1) \end{array}$$

according to formula

$$\Rightarrow \begin{bmatrix} \boxed{1} & -2 & 1 & 0 \\ 0 & \boxed{2} & -8 & 8 \\ 0 & 0 & \boxed{1} & 3 \end{bmatrix}; \begin{array}{l} R_3 + \frac{3}{2} R_2 \\ -3 \pm \frac{3}{2} (2) \end{array}$$

BASIC VARIABLE:-

"column one contains a pivot position.

so the variable corresponding to column one (x_1) is called....."

"column two also contains

so variable corresponding to this column (x_2) is....."

"This imply that x_3 is also basic."

"If there is no basic variable or leading entry in a column so that variable is FREE

VARIABLE."

NOTE:-

• If all the variables in a given system are basic than given system has unique soln.

• If there exists at least one free variable

than given system has infinitely many solutions.

• If the right most column of the augmented matrix is pivot than has no soln.

All variables in prev. ex. are basic, so system has unique soln.

$$x_1 - 2x_2 + x_3 = 0 \quad \text{--- (i)}$$

$$2x_2 - 8x_3 = 8 \quad \text{--- (ii)}$$

$$x_3 = 3 \quad \text{--- (iii)}$$

$$(x_1, x_2, x_3) = (29, 16, 3)$$

EQUIVALENT:-

“Two systems are called equivalent and their solns. are same.”

Ex 2

$$x_2 - 4x_3 = 8$$

$$2x_1 - 3x_2 + 2x_3 = 1$$

$$5x_1 - 8x_2 + 7x_3 = 1$$

ol

$$\begin{bmatrix} 0 & 1 & -4 & 8 \\ 2 & -3 & 2 & 1 \\ 5 & -8 & 7 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 5 & -8 & 7 & 1 \end{bmatrix} \quad \begin{array}{l} R_{12} = \text{interchange} \\ R_3 \pm \square R_1 \\ 5 - (\frac{5}{2})2 \end{array}$$

$$\sim \begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & -1 & 2 & -3/2 \end{bmatrix} \quad \begin{array}{l} R_3 \pm \square R_2 \\ -\frac{1}{2} + \frac{1}{2}(2) \end{array}$$

$$\sim \begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5/2 \end{bmatrix}$$

This has no soln. As right most column of augmented matrix is pivot.

Ex 3

$$\begin{aligned}x_1 + 5x_2 - 3x_3 &= -4 \\ -x_1 - 4x_2 + x_3 &= 3 \\ -2x_1 - 7x_2 &= 5\end{aligned}$$

Sol

$$\begin{bmatrix} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Here x_1 & x_2 are basic variables while x_3 is free variable. So given system has infinitely many solns

$$x_1 + 5x_2 - 3x_3 = -4$$

$$x_2 - 2x_3 = -1$$

write basic variables in the form of free variable

$$x_1 + 5x_2 = -4 + 3x_3$$

$$x_2 = -1 + 2x_3$$

$$x + y - 4 = 0$$

$$y = 4 - x$$

value y in the form of x on in terms of x .

- we can change the values of x variables because it is a free variable.
- But we cannot change the value of basic variable
- The value of free variables change/vary because it has infinite many soln.
- so, corresponding to have any value of x_3 , we have a soln. (Ex. answer). as If we put value in above eq. as $x_3 = 0, x_3 = 1 / x_3 = 2 / x_3 = 4 \dots \dots$

Def

⇒ "A given system is consistent if it has unique soln. or infinitely many soln."

⇒ "..... inconsistent no soln."

ROW OPERATIONS:-

- $R_3 \pm \square R_1$
- $R_{12} \rightarrow$ inter change.
- αR_2
any non zero constants

$2, -1, -6 \text{ \& } 3 = \text{constants}$
 $x_1 \text{ \& } x_2 = \text{variables}$
 $h \text{ \& } K = \text{parameters}$

Ques

$$\begin{aligned} 2x_1 - x_2 &= h \\ -6x_1 + 3x_2 &= K \end{aligned}$$

$$\begin{bmatrix} 2 & -1 & h \\ -6 & 3 & K \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & h \\ 0 & 0 & K+3h \end{bmatrix}$$

if $K+3h \neq 0$
 $\Rightarrow$ system is inconsistent.
if $K+3h = 0$
 $\Rightarrow$ system is consistent.

Lec 15:-

Sec 1.3:- VECTORS:-

"A matrix which has only one column."

Ex $\begin{bmatrix} 1 \\ 3 \end{bmatrix} \in \mathbb{R}^2$
 $\vec{a} = \hat{i} + 3\hat{j}$

$\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} \in \mathbb{R}^3$
 $\vec{b} = \hat{i} + 3\hat{j} + 0\hat{k}$

$\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$
2D 3D

EQUAL VECTORS:-

"Two vectors are said to be equal if their corresponding entries are same."

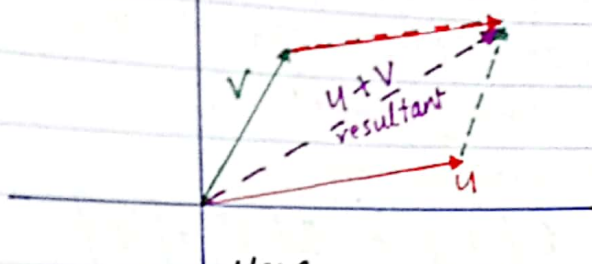
Ex:- if $U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $V = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ $U \text{ \& } V$ are equal vectors
if $u_1 = v_1$, $u_2 = v_2$

ADDITION OF VECTORS:-

if $U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $V = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

$$U + V = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

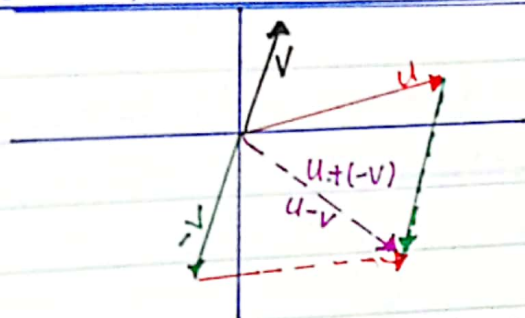
GEOMETRICALLY ADDITION OF VECTORS:-



$\Rightarrow U+V = \text{resultant}$
always lie on the
same plane in which
 U & V lies.

- Here $U+V$ lies on the same plane on which U and V lies.

GEOMETRICALLY SUBTRACTION OF VECTORS:-



$$U-V = ?$$

$$U+(-V) = ?$$

PARALLELOGRAM RULES OF ADDITION:-

- If U and V are two vectors that belongs to $\mathbb{R}^2$ (2D) then $U+V$ forms the fourth vertex of the parallelogram whose other vertices are U, V & O .

Ex. 1 if $W = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$, $U = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $V = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

Is W has linear combination with U & V ? If yes, find that combination.

W has linear if there exists scalars x_1 & x_2 such that

$$x_1 U + x_2 V = W$$

$$x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 1 \end{bmatrix} = W = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$x_1 + 3x_2 = 4$$

$$2x_1 + x_2 = 3$$

$$\begin{bmatrix} 1 & 3 & 4 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 4 \\ 0 & -5 & -5 \end{bmatrix} R_2 - 2R_1$$

$$x_1 + 3x_2 = 4$$

$$-5x_2 = -5$$

$$x_2 = 1$$

$$x_1 = 1$$

As x_1 & x_2 exists so w has L.C with u & v .

$$1u + 1v = w$$

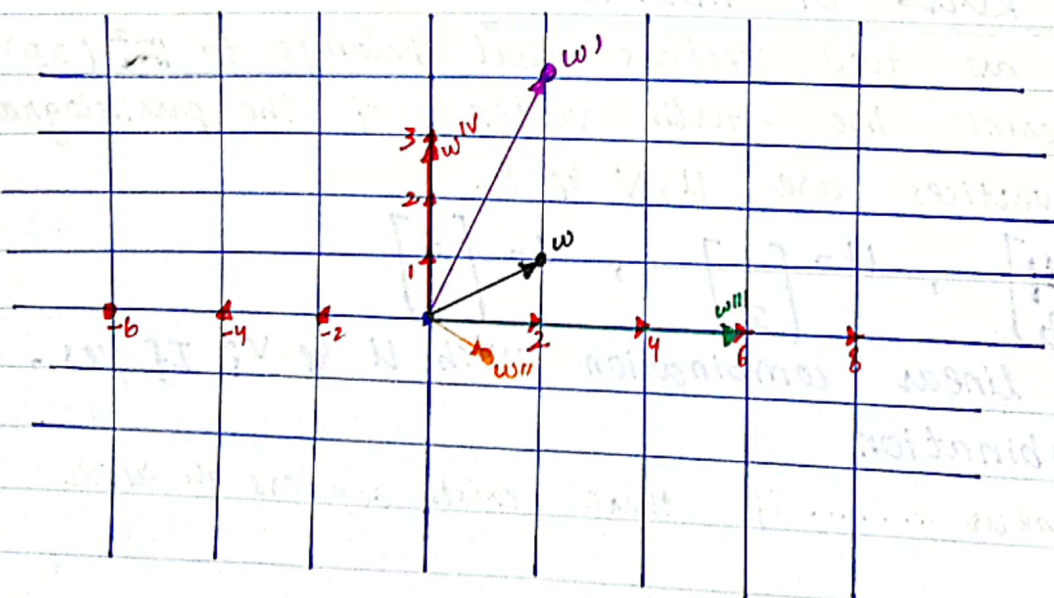
Verification.

$$1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

Ex 2

$$u = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



$$1u + 1v = w \quad \text{--- (1)}$$

$$1u + 3v = w' \quad \text{--- (2)}$$

$$\frac{1}{2}u - \frac{1}{2}v = w'' \quad \text{--- (3)}$$

$$3u + 0v = w'''$$

$$0u + 3v = w''''$$

from eq. we has L.C with u & v why? because there exists scalars like $x_1 = 1, x_2 = 1$.

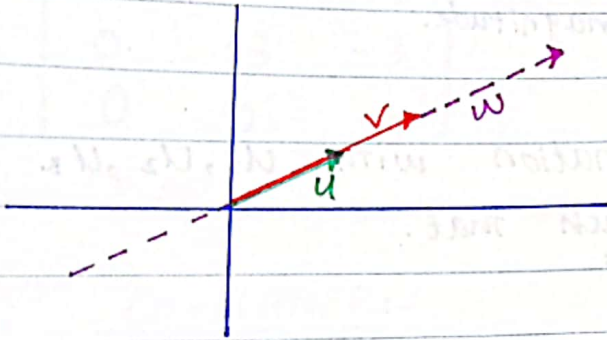
Ex 3

if $u = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$, $w = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$

w has linear combination with u & v . as

$$1u + 1v = w$$

because there exists scalar like $x_1=1, x_2=1$



Mathematically,

w has L.C with u & v if there exists scalars.

x_1 & x_2 $\exists$

$$x_1 u + x_2 v = w$$

$$x_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

$$2x_1 + 4x_2 = 6$$

$$x_1 + 2x_2 = 3$$

$$\begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix} R_{12} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 + 2x_2 = 3$$

$$x_1 = 3 - 2x_2$$

if $x_2 = 1$, $x_1 = 1$

As x_1 & x_2 exists so w

has L.C with u & v

and this L.C is $1u + 1v = w$

Ex 4

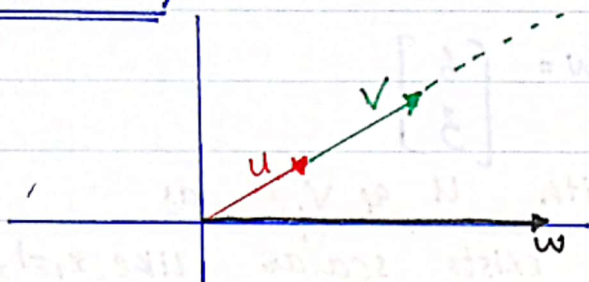
if $u = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$, $w = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$

$$\dots \begin{bmatrix} 2 & 4 & 5 \\ 1 & 2 & 0 \end{bmatrix} R_{12} \Rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 5 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

so, it has no soln. and x_1 & x_2 do not exist.

so, w has no L.C with u & v .

GRAPHICALLY,



There is no relation because the multiplication of any no. with scalars doesn't change the direction of vectors. Only changes magnitude.

DEFINITION:-

w has linear combination with u_1, u_2, u_3 .

If there exists scalars α such that.
like x_1, x_2, x_3

$$x_1 u_1 + x_2 u_2 + x_3 u_3 = w$$

Ex Is v has l.c with v_1, v_2, v_3 ? If yes, find that combination.

$$v = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}, v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

If v has l.c.....

$$x_1 v_1 + x_2 v_2 + x_3 v_3 = v.$$

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}$$

$$x_1 + 2x_2 - x_3 = 2$$

$$2x_1 + x_2 + x_3 = 4$$

$$3x_1 + 0x_2 + x_3 = 4$$

$$\begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & -3 & 3 & 0 \\ 0 & 0 & -2 & -2 \end{bmatrix}$$

So, soln. exists.

$$x_1 + 2x_2 - x_3 = 2$$

$$-3x_2 + 3x_3 = 0$$

$$-2x_3 = -2$$

$$\begin{aligned} x_3 &= 1 \\ x_2 &= 1 \\ x_1 &= 1 \end{aligned}$$

$$1v_1 + 1v_2 + 1v_3 = v$$



$$V = \begin{bmatrix} 3 \\ 3 \\ -1 \end{bmatrix}, V_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, V_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad V = \begin{bmatrix} -6 \\ -6 \\ -6 \end{bmatrix}, V_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, V_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -3 \\ 0 & 0 & -4 \end{bmatrix} \quad \Rightarrow \begin{bmatrix} 1 & 2 & -6 \\ 0 & -3 & 6 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow$ No soln. $\Rightarrow$ Have soln.

Extra Ex

CO-PLANER:-

$$V_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, V_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, V_3 = \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix}$$

These vectors are coplaner if V_3 has L.C with V_1, V_2

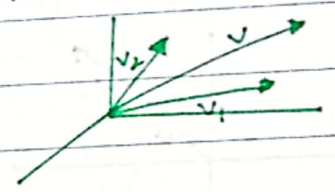
$$x_1 V_1 + x_2 V_2 = V_3$$

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix}$$

$$x_1 = 1$$

$$x_2 = 1$$

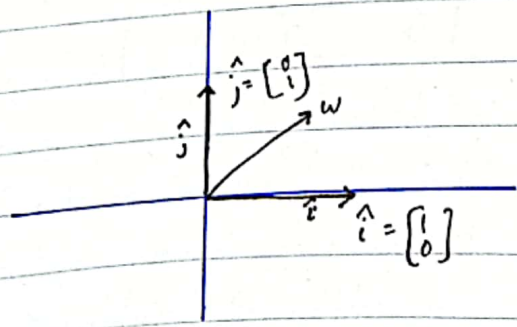
As x_1 & x_2 exists so V_3 has L.C with V_1 & V_2 . So, V_1, V_2, V_3 are co-planer.



SPAN:-

"The collection of all L.Cs is called span."

• span of $\{u, v\}$ means what is generated by u & v .



$$\text{span} \{ \hat{i}, \hat{j} \} = \mathbb{R}^2$$

$$\text{span} [\hat{i}] = \text{Line in } \mathbb{R}^2$$

THEOREM:- This is applicable only on $\downarrow$ matrix.

Let A be an $m \times n$ co-efficient matrix. Then the following statements are logically equivalent (Means either they are all true or all are false.).

a span of columns if A is $\mathbb{R}^m$.

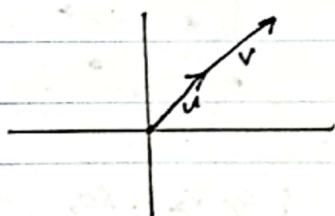
b A has pivot position in every row.

Ex:1 if $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ $\text{span}\{u, v\} = ?$

sol $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$

Here 1st row is pivot but 2nd is not pivot so A has not pivot position in every row.

$\Rightarrow$ so span of columns of A is not $\mathbb{R}^2$



$\text{span}\{u, v\} \neq \mathbb{R}^2$

$\text{span}\{u, v\} = \text{line in } \mathbb{R}^2$

Ex:2 $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$

A has pivot position in every row

$\Rightarrow$ so span of columns of A is $\mathbb{R}^m \rightarrow \mathbb{R}^2$

$\text{span}\{u, v\} = \mathbb{R}^2$

If 2 pivot = line
If 2 pivot = plane
If 3 pivot = $\mathbb{R}^3$



Ex. 3:-

if $u = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$A = \begin{bmatrix} \boxed{2} & 0 \\ 0 & \boxed{1} \end{bmatrix}$$

A has so span $\mathbb{R}^2$, $\text{span}\{u, v\} = \mathbb{R}^2$

Ex. 4:-

if $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$, $w = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$

$\text{span}\{u, v, w\}$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} \boxed{1} & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

A has not so span is not $\mathbb{R}^2$.

$\text{span}\{u, v, w\} = \text{line in } \mathbb{R}^2$

Ex. 5:-

$u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, $w = \begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix}$

$\text{span}\{u, v, w\} = ?$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -7 \\ 0 & 0 & 10 \end{bmatrix}$$

A has so span of $\mathbb{R}^3$, $\text{span}\{u, v, w\} = \mathbb{R}^3$.

Result:- These 3 vectors are not coplanar so they are plane in $\mathbb{R}^3$.

Ex. 6

$u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, $w = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$

$$A = \begin{bmatrix} \boxed{1} & 2 & 3 \\ 0 & \boxed{-3} & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

A has not so span is not $\mathbb{R}^3$

$\text{span} \neq \mathbb{R}^3$

$\text{span}\{u, v, w\} = \text{plane in } \mathbb{R}^3$.

Because they are coplanar.

Ex-7 $u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ -2 \\ -3 \end{bmatrix}$, $w = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$ $\text{span}\{u, v, w\} = ?$

$$A = \begin{bmatrix} 1 & -1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

A has not So span is not $\mathbb{R}^3$. This is line in $\mathbb{R}^3$.
Because they are multiple of each other that's why they are line in $\mathbb{R}^3$. $\downarrow$ collinear

let $a = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$, $a_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}$, $b = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}$ prove that $\text{span}\{a_1, a_2\} = \text{plane in } \mathbb{R}^3$

$$A = \begin{bmatrix} 1 & 5 \\ -2 & -13 \\ 3 & -3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 5 \\ 0 & -3 \\ 0 & 0 \end{bmatrix}$$

As every row is not pivot, so $\text{span}\{a_1, a_2\} \neq \mathbb{R}^3$
 $\text{span}\{a_1, a_2\} = \text{plane in } \mathbb{R}^3$
 b has l.e with a_1 & a_2

$$x_1 a_1 + x_2 a_2 = b$$

$$x_1 \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}$$

$$x_1 + 5x_2 = -3$$

$$-2x_1 - 13x_2 = 8$$

$$3x_1 - 3x_2 = 1$$

$$\begin{bmatrix} 1 & 5 & -3 \\ -2 & -13 & 8 \\ 3 & -3 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & 0 & -2 \end{bmatrix} \Rightarrow \text{No soln.}$$

$\Rightarrow x_1, x_2$ not exist.
 $\Rightarrow b$ has no
 $\Rightarrow b$ not belongs to $\text{span}\{a_1, a_2\}$

$$a_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}, \quad a_3 = \begin{bmatrix} 4 \\ -11 \\ -6 \end{bmatrix}$$

a) span $\{a_1, a_2\}$ = plane in $\mathbb{R}^3$

b) span $\{a_1, a_2, a_3\}$ = plane in $\mathbb{R}^3$.

c) prove that they are in same plane:

$$\begin{bmatrix} 1 & 5 & 4 \\ -2 & -13 & -11 \\ 3 & -3 & -6 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 5 & 4 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

co-efficient matrix

Both are in same plane if a_3 lies in plane of a_1 & a_2

$$x_1 a_1 + x_2 a_2 = a_3$$

$$x_1 \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix} = \begin{bmatrix} 4 \\ -11 \\ -6 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 5 & 4 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

unique soln.

augmented.

Q: $0x_1 + 3x_2 - 6x_3 + 6x_4 + 4x_5 = -5$

$$3x_1 - 7x_2 + 8x_3 - 5x_4 + 8x_5 = 9$$

$$3x_1 - 9x_2 + 12x_3 - 9x_4 + 6x_5 = 15$$

solve this system & write its parametric form of soln.

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \\ 0 & 0 & 0 & 0 & 21 & 81 \end{bmatrix}$$

Here this system has infinitely many solns.

because x_3 & x_4 are free variables.

On right side there is
always free variable not
basic variable.



$$3x_1 - 7x_2 + 8x_3 - 5x_4 + 8x_5 = 9$$

$$3x_2 - 6x_3 + 6x_4 + 4x_5 = -5 \Rightarrow$$

$$\frac{2}{3} x_5 = 8/3$$

$$x_5 = 4, \quad 3x_1 - 7x_2 = -23 - 8x_3 + 5x_4$$

$$3x_2 = -21 + 6x_3 - 6x_4$$

$$x_2 = -7 + 2x_3 - 2x_4$$

putting values = $3x_1 - 7(-7 + 2x_3 - 2x_4) = -23 - 8x_3 + 5x_4$

$$3x_1 = -72 + 6x_3 - 9x_4$$

$$x_1 = -24 + 2x_3 - 3x_4$$

$$x_1 = -24 + 2x_3 - 3x_4, \quad x_3 = x_3, \quad x_5 = 4.$$

$$x_2 = -7 + 2x_3 - 2x_4, \quad x_4 = x_4$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -24 + 2x_3 - 3x_4 \\ -7 + 2x_3 - 2x_4 \\ x_3 \\ x_4 \\ 4 \end{bmatrix} \Rightarrow \begin{bmatrix} -24 + 2x_3 - 3x_4 \\ -7 + 2x_3 - 2x_4 \\ 0 + 1x_3 + 0x_4 \\ 0 + 0x_3 + 1x_4 \\ 4 + 0x_3 + 0x_4 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -24 \\ -7 \\ 0 \\ 0 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -3 \\ -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

This is the parametric
form of soln.

if $x_3 = 0, x_4 = 0$, $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -24 \\ -7 \\ 0 \\ 0 \\ 4 \end{bmatrix}$

if $x_3 = 0, x_4 = 1$ = $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -22 \\ -5 \\ 0 \\ 1 \\ 4 \end{bmatrix}$

Any system of linear eq. $Ax=b$ is non-homogenous if $b \neq 0$ and homogenous if $b=0$.

Ex

$$3x_1 + 2x_2 - x_3 = 5$$

$$-x_1 + x_2 - x_3 = 0$$

$$3x_1 + x_2 - x_3 = 0$$

This is non-homogenous. Linear system.
This have 3 possibilities of soln. (uni, infinite, no)

$$3x_1 + 2x_2 - x_3 = 0$$

$$-x_1 + x_2 - x_3 = 0$$

$$3x_1 + x_2 - x_3 = 0$$

This is homogenous Linear system.
This have 2 po... (uni, infinite).

$\Rightarrow$ There is no possibility that homogenous linear system has no solution because (all) right most columns of augmented matrix can never be pivot.

Sec 1.7
Ex 1

$$0x_1 + 1x_2 + 4x_3 = 0$$

$$x_1 + 2x_2 - x_3 = 0$$

$$5x_1 + 8x_2 + 0x_3 = 0$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

If $Ax=0$ has unique soln. than this unique soln. is $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. This is called trivial (obvious) soln. $\Rightarrow$ vectors are independent

Ex 2

$$x_1 + 4x_2 + 2x_3 = 0$$

$$2x_1 + 5x_2 + x_3 = 0$$

$$3x_1 + 6x_2 + 0x_3 = 0$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

So, This is the parametric form of the soln.

If $Ax=0$ has infinitely many solns. than this system has non-trivial soln. $\Rightarrow$ vectors are dependent.



Ex

let $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, $v_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ Determine if the set which consist of three

vectors is dependent? If than find relation?

a// given set of vectors are linearly dependent if $Ax=0$ as non-trivial soln. Given set of vectors is linearly independ...

if $Ax=0$ as trivial soln.

$$Ax=0 \Rightarrow \begin{bmatrix} 1 & 4 & 2 \\ 2 & 5 & 1 \\ 3 & 6 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

we write x variable according to columns of A

$$x_1 + 4x_2 + 2x_3 = 0$$

$$2x_1 + 5x_2 + x_3 = 0$$

$$3x_1 + 6x_2 + 0x_3 = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

since $Ax=0$ has infinitely many solns. & non-trivial soln

b// $x_1 + 4x_2 + 2x_3 = 0$

$$-3x_2 - 3x_3 = 0$$

$$\Rightarrow x_2 = -x_3$$

$$x_3 = x_3$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\Rightarrow x_1 + 4(-x_3) + 2x_3 = 0$$

$$x_1 = 2x_3$$

The relation is $2v_1 - v_2 + v_3 = 0$

Note:-

1. If the given set has more vectors than there no. of entries than the set is ^{always} linearly dependent. $\{v_1, v_2, v_3, v_4\}$

2. If a set contains '0' vector than the set is always

Linearly dependent as $\{v_1, v_2, 0, v_4\}$

3. Is single non zero vectors are always linearly independent.

4. A single zero vector is always linearly dependent.

$$xv_1 = 0 \Rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} x = 0 \Rightarrow x = 1, 2, 3, 4, \dots$$

Ex

$$10x_1 - 3x_2 - 2x_3 = 0$$

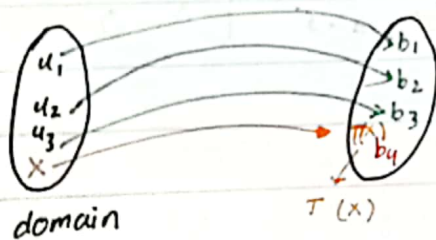
$$\begin{bmatrix} 10 & -3 & -2 & 0 \end{bmatrix} \Rightarrow 10x_1 = 3x_2 + 2x_3 \Rightarrow x_1 = 0.3x_2 + 0.2x_3$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.3x_2 + 0.2x_3 \\ x_2 \\ x_3 \end{bmatrix} \Rightarrow \begin{bmatrix} 0.3x_2 + 0.2x_3 \\ 1x_2 + 0x_3 \\ 0x_2 + 1x_3 \end{bmatrix} \Rightarrow x_2 \begin{bmatrix} 0.3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0.2 \\ 0 \\ 1 \end{bmatrix}$$

LINEAR TRANSFORMATIONS:-

(matrices are the func. of/in)
L.A

A transformation (func. or mapping) $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ that assigns to each vectors $x \in \mathbb{R}^n$ to a vector $T(x) \in \mathbb{R}^m$.



Range + Codomain.

not range, but codomain.

because its image is not made in codomain from domain.

Domain, Codomain & Range:-

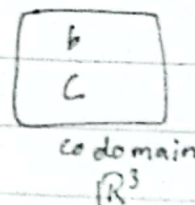
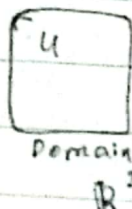
The set of all input vectors belongs to the domain & set of all output vectors called codomain. "Range consists of all those vectors which are the images of some vectors. Range is the image of the any vector (کسی نہ کسی)"

Ex. 1 let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $c = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$

Find order of transformation:-

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} * \\ * \end{bmatrix} = \begin{bmatrix} * \\ * \\ * \end{bmatrix}$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$$



Find image of U :-

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$

$\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ is transformed into $\begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$ under A .



Find a vector whose image is b :-

$$AX = b$$

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x_1 - 3x_2 &= 3 \\ 3x_1 + 5x_2 &= 2 \\ -x_1 + 7x_2 &= -5 \end{aligned} \Rightarrow \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix} \begin{aligned} & \\ & \\ & \end{aligned}$$

$\begin{aligned} &R_2 - 3R_1 \\ &R_3 - R_1 \\ &R_2 + 14R_1 \\ &R_3 - 4R_2 \end{aligned}$

$$\begin{aligned} x_1 - 3x_2 &= 3 \\ x_2 &= -1/2 \end{aligned} \Rightarrow \begin{aligned} x_1 + 3/2 &= 3 \\ x_1 &= 3/2 \end{aligned}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}$$

is a vector whose image is b .

Is $c \in \text{Range}$?

$$\begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 10 \end{bmatrix}$$

As right most coln. is pivot which indicates that above system has no soln. So, there ~~don't~~ does not exist x whose image is c . So c doesn't belong to range. of course $c \in \text{codomain}$.

- q
 a// Codomain = range. (improper subset).
 b// Codomain $\neq$ range (range is proper subset of codomain)
 How to prove?

$\Rightarrow$ If every row of coefficient matrix is pivot then Codomain = range.

$\Rightarrow$ If every row not pivot then Codomain $\neq$ range

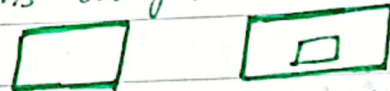
Q Lets us consider a transformation:

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 2 & 4 \\ 3 & 1 & 4 \end{bmatrix}$$

a// Is Codomain = range or not?

$$\sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

As every row is not pivot so $cd \neq range$



b// Find a vector that lies in Codomain but not in range.
 let b be a vector that lies in Codomain but doesn't belong to range.

$$Ax = b$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 2 & 2 & 4 \\ 3 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -1 & 0 & b_1 \\ 2 & 2 & 4 & b_2 \\ 3 & 1 & 4 & b_3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -1 & 0 & b_1 \\ 0 & 4 & 4 & b_2 - 2b_1 \\ 0 & 4 & 4 & b_3 - 3b_1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 0 & b_1 \\ 0 & 4 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 - b_1 - b_2 \end{bmatrix}$$

above soln. system has no soln. if $-b_1 - b_2 + b_3 \neq 0$

As, $b_1 = 1, b_2 = 2, b_3 = 1$

So $b = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ is a vector that lies in Codomain but not in range.



Find a vector that lies in codomain and also in range.
system has soln. if $c_3 - c_1 - c_2 = 0$

$$b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

so $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is a vector that lies in codomain and also in range.

Q $A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 2 & 4 \\ 3 & 1 & -4 \end{bmatrix}$

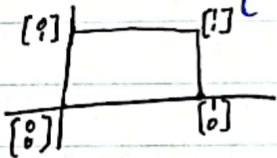
$\sim \begin{bmatrix} \boxed{1} & -1 & 0 \\ 0 & \boxed{4} & 4 \\ 0 & 0 & \boxed{-8} \end{bmatrix}$

codomain = range.

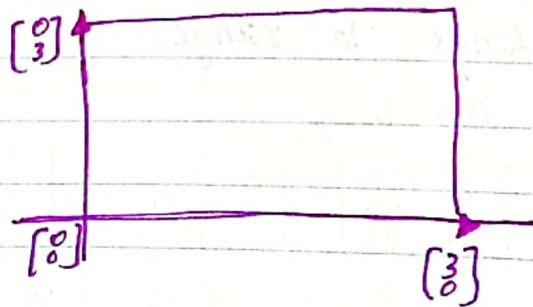
As codomain = range, so there doesn't exist any vector that lies in codomain but not in range.

Properties of matrices:-

Q $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$



apply it on a unit square.



$$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

⇒ so given transformation increases the ^{geometry/vector} size of $\uparrow$ /unit square 3-times.

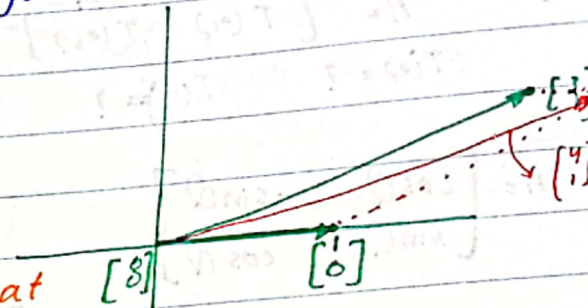
• $A = \begin{bmatrix} \delta & 0 \\ 0 & \delta \end{bmatrix}$ is called dilation transformation if $\delta > 1$ and called contraction transformation if $0 < \delta < 1$.

• $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ res 2-times.

Ex 2:-

Also let $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$ Apply given transformation on a unit square. Also describe the effect of this transformation.

$$\begin{aligned} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \end{bmatrix} &\rightarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} & \begin{bmatrix} 1 \\ 0 \end{bmatrix} &\rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} &= \begin{bmatrix} 3 \\ 1 \end{bmatrix} & \begin{bmatrix} 0 \\ 1 \end{bmatrix} &\rightarrow \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} 4 \\ 1 \end{bmatrix} & \begin{bmatrix} 1 \\ 1 \end{bmatrix} &\rightarrow \begin{bmatrix} 4 \\ 1 \end{bmatrix} \end{aligned}$$

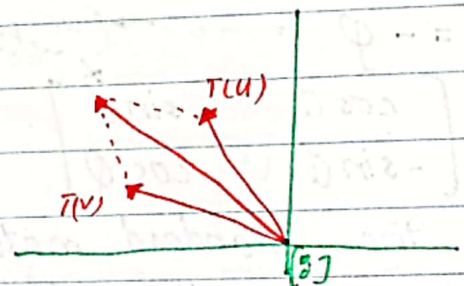


So this transformation has property that it transforms a square into parallelogram.

Ex Let T be a transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x) = Ax$ where $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ Apply this transformation on a parallelogram formed by

$$u = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, u+v=w = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

$$\begin{aligned} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} &= \begin{bmatrix} -1 \\ 4 \end{bmatrix} \\ \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} &= \begin{bmatrix} -3 \\ 2 \end{bmatrix} \\ \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix} &= \begin{bmatrix} -4 \\ 6 \end{bmatrix} \end{aligned}$$



so given transformation has the property that it rotates every vector through an angle of 90° in counter-clockwise rotation about origin.

Ex let T be a transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ & $T(e_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix}$ $T(e_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix}$ Find the standard matrix that transforms $\begin{bmatrix} 4 \\ 0 \end{bmatrix}$ into $\begin{bmatrix} -5 \\ -7 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ into $\begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix}$

$$A = [T(e_1) \ T(e_2)]$$

$$A = \begin{bmatrix} 5 & -3 \\ -7 & 8 \\ 2 & 0 \end{bmatrix}$$

$$T(x) = \begin{bmatrix} 5 & -3 \\ -7 & 8 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$T(x) = Ax.$$

• cartesian coordinates
• polar

$$x = r \cos \theta$$

$$y = r \sin \theta$$

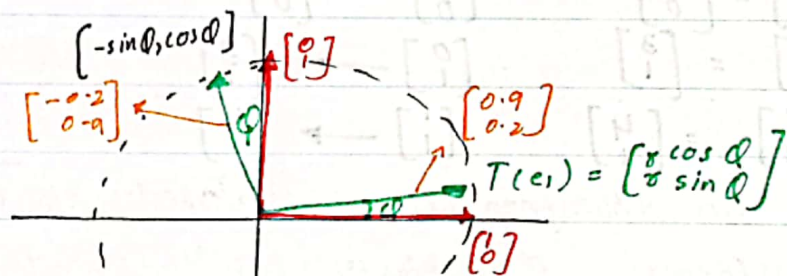


Find the standard matrix that rotates every vector through an angle of ϕ in counter-clockwise direction about origin

sol/ $A = [T(e_1) \ T(e_2)]$

$$T(e_1) = ? \quad T(e_2) = ?$$

$$A = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$



This matrix has the property that it rotates every vector through an angle ϕ in counter-clockwise direction about origin

If $\phi = 90^\circ$ then $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ $\begin{bmatrix} x & y \\ y & x \end{bmatrix}$

For clockwise direction:-

$\phi = -\phi = -60^\circ \rightarrow$ (we put 60 in below matrix not -60° but -60° in above matrix A)
 $A = \begin{bmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{bmatrix}$ because $\cos$ is even func.
 so, $-\cos \phi = \cos \phi$

• +ve sign with angle indicates anticlockwise direction and -ve = clockwise

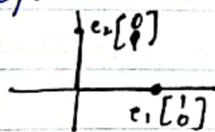
• $\phi = 45^\circ \Rightarrow$ anticlockwise

• $\phi = -45^\circ \Rightarrow$ clockwise

• sign shows only direction, not included in putting in func.

Find the standard matrix that gives:

Reflection through x_1 -axis.



$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

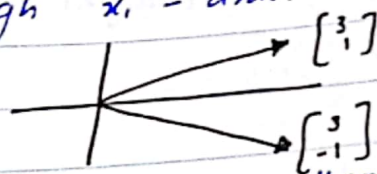
$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

This matrix has a property that it reflects every vector through x_1 -axis.

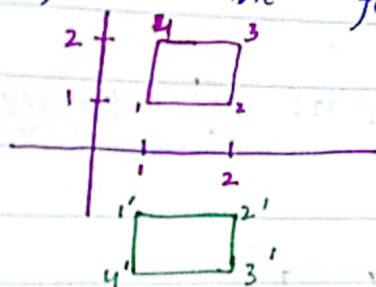
Ex

- Reflect the vector $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ through x_1 -axis.

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$



- Reflect the following unit squares



- Reflect vector through x_2 -axis.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Reflect vector $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ through x_2 -axis.

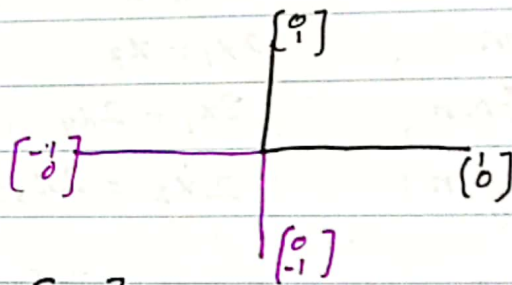
$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

- Reflect through origin.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$



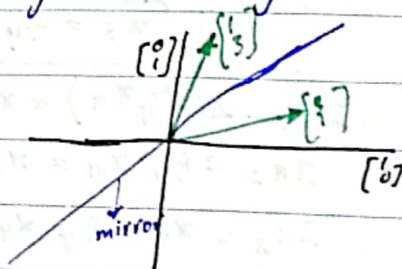
- Reflect $\begin{bmatrix} 3 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$

- Reflection through straight line $x_2 = x_1$.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$



$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

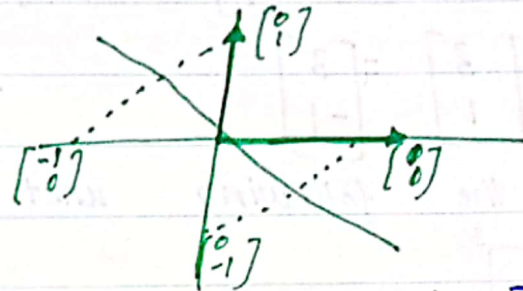
e

Reflection through straight line $x_2 = -x_1$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$



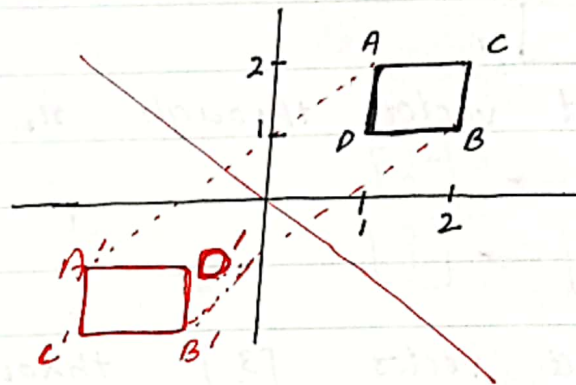
Q Reflect the given unit square (in part a) through straight line $x_2 = -x_1$.

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \rightarrow \begin{bmatrix} -2 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 2 \end{bmatrix} \rightarrow \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$



Propan (C_3H_8) combines with oxygen (O_2) to form CO_2 & H_2O .



Carbon, $3x_1 = x_3$

Hydrogen, $8x_1 = 2x_4$

Oxygen, $2x_2 = 2x_3 + x_4$

$$3x_1 + 0x_2 - 1x_3 + 0x_4 = 0$$

$$8x_1 + 0x_2 + 0x_3 - 2x_4 = 0$$

$$0x_1 + 2x_2 - 2x_3 - x_4 = 0$$

$$\begin{bmatrix} 3 & 0 & -1 & 0 & 0 \\ 8 & 0 & 0 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1/3 & 0 & 0 \\ 0 & 2 & -2 & -1 & 0 \\ 0 & 0 & 8/3 & -2 & 0 \end{bmatrix}$$

$$\Rightarrow x_1 - \frac{1}{3}x_3 = 0$$

$$\Rightarrow 2x_2 - 2x_3 - x_4 = 0$$

$$\Rightarrow \frac{8}{3}x_3 - 2x_4 = 0$$

$$x_1 - \frac{1}{3}x_3 = 0$$

$$2x_2 - 2x_3 = x_4$$

$$x_3 = \frac{3}{4}x_4$$

$$2x_2 - 2\left(\frac{3}{4}x_4\right) = x_4$$

$$2x_2 - \frac{6}{4}x_4 = x_4$$

$$2x_2 = x_4 + \frac{6}{4}x_4$$

$$= \frac{10}{4}x_4$$

$$x_2 = \frac{5}{4}x_4$$

$$x_1 - \frac{1}{3}\left(\frac{3}{4}x_4\right) = 0$$

$$x_1 = \frac{1}{4}x_4$$

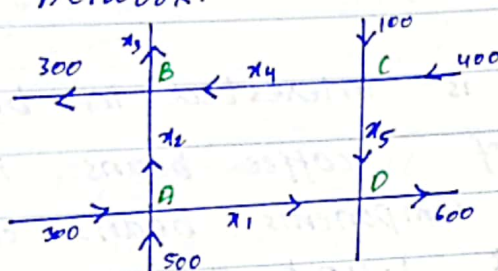
$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \frac{1}{4}x_4 \\ \frac{5}{4}x_4 \\ \frac{3}{4}x_4 \\ x_4 \end{bmatrix}$$

if $x_4 = 4 \Rightarrow x_1 = 1, x_3 = 3$
 $x_2 = 5, x_5 = 4$

The eq. is $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$.

Ex.02:-

The network in the following figure shows the traffic flow over the given several one way streets. Determine the general flow pattern for the given network.



A, B, C, D = conjunction points

Intersection	Flow in	Flow out
A	300 + 500	$x_1 + x_2$
B	$x_2 + x_4$	$x_3 + 300$
C	400 + 100	$x_5 + x_4$
D	$x_1 + x_5$	600

Total flow in = Total flow out } this all is outside the box like outside conjunctions.
 $100 + 400 + 500 + 300 = 300 + x_3 + 600$
 $x_3 = 400$

$$\Rightarrow \begin{aligned} x_1 + x_2 &= 800 \\ x_2 + x_4 &= x_3 + 300 \\ x_5 + x_4 &= 500 \\ x_1 + x_5 &= 600 \\ x_3 &= 400 \end{aligned}$$

$$\begin{aligned} x_1 + x_2 + 0x_3 + 0x_4 + 0x_5 &= 800 \\ 0x_1 + x_2 - x_3 + x_4 + 0x_5 &= 300 \\ 0x_1 + 0x_2 + 0x_3 + 1x_4 + 1x_5 &= 500 \\ x_1 + 0x_2 + 0x_3 + 0x_4 + 1x_5 &= 600 \\ 0x_1 + 0x_2 + 1x_3 + 0x_4 + 0x_5 &= 400 \end{aligned}$$

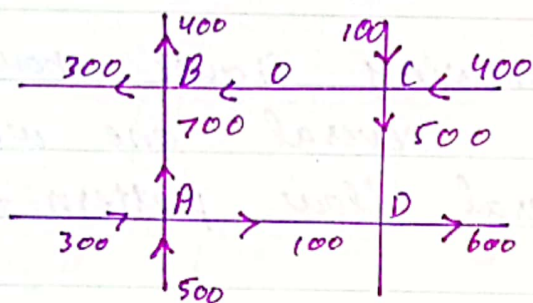
[echelon form] $\rightarrow$

$$\begin{aligned} x_1 &= 600 - x_5 \\ x_2 &= 200 + x_5 \\ x_3 &= 400 \\ x_4 &= 500 - x_5 \\ x_5 &\text{ is free} \end{aligned}$$

Here x_5 is free and we can assume any value of $x_5 \leq 500$.

let $x_5 = 500$ than

$$\begin{aligned} x_1 &= 100 \\ x_2 &= 700 \\ x_3 &= 400 \\ x_4 &= 0 \end{aligned}$$



2. A coffee manufacturer is interested in blending three different types of coffee beans into a final coffee blend. The 3 components beans cost the manufacturer 1.20\$, 1.60\$, 1.40\$. The manufacturer wants to blend a batch of 40,000\$ of coffee and has a coffee purchasing budget of 57,600\$. In blending the coffee, this is called modeling of mathematics. One restriction is that the amount used for/of component 2 should be twice (2) to that of component 1. for better flavour. Determine whether there is a combination of 3 components which will lead to the final blend consisting of 40,000 pound cost of 57,600\$ and satisfying the blending restriction.

x_1 = weight of component 1

x_2 = " " " 2

x_3 = " " " 3

$$x_1 + x_2 + x_3 = 40,000$$

$$2x_1 = x_2$$

$$1.20x_1 + 1.60x_2 + 1.40x_3 = 57,600$$

$$x_1 + x_2 + x_3 = 40,000$$

$$1.20x_1 + 1.60x_2 + 1.40x_3 = 57,600$$

$$2x_1 - x_2 + 0x_3 = 0$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 & 40000 \\ 1.20 & 1.60 & 1.40 & 57600 \\ 2 & -1 & 0 & 0 \end{bmatrix}$$

Find the standard matrix:- that gives
a// Projection onto x_1 -axis:-

$$A = [T(e_1) \quad T(e_2)]$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

This matrix has the
projection onto x_1 -axis
property.

b// Projection onto x_2 -axis:-

$$A = [T(e_1) \quad T(e_2)]$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Previous Application Part:-

$$\begin{bmatrix} 1 & 1 & 1 & 40000 \\ 0 & 0.4 & 0.2 & 9600 \\ 0 & 0 & -1/2 & -8000 \end{bmatrix} \Rightarrow$$

$$x_1 = 8000$$

$$x_2 = 16,000$$

$$x_3 = 16,000$$

Q Consider the following letter N whose eight points
or vertices 1, 2, 3, 4, 5, 6, 7 and 8 are determined
by following vectors respectively,

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \end{bmatrix}, \begin{bmatrix} 6.42 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ 8 \end{bmatrix}, \begin{bmatrix} 5.5 \\ 8 \end{bmatrix}, \begin{bmatrix} 5.5 \\ 1.58 \end{bmatrix}, \begin{bmatrix} 0 \\ 8 \end{bmatrix}$$

Apply the transformation
new image of N and
transformation.

$A = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix}$ to find
also describe effect of given

$$A_1 = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 6.42 \\ 0 \end{bmatrix} = \begin{bmatrix} 6.42 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 2.105 \\ 6.42 \end{bmatrix}$$

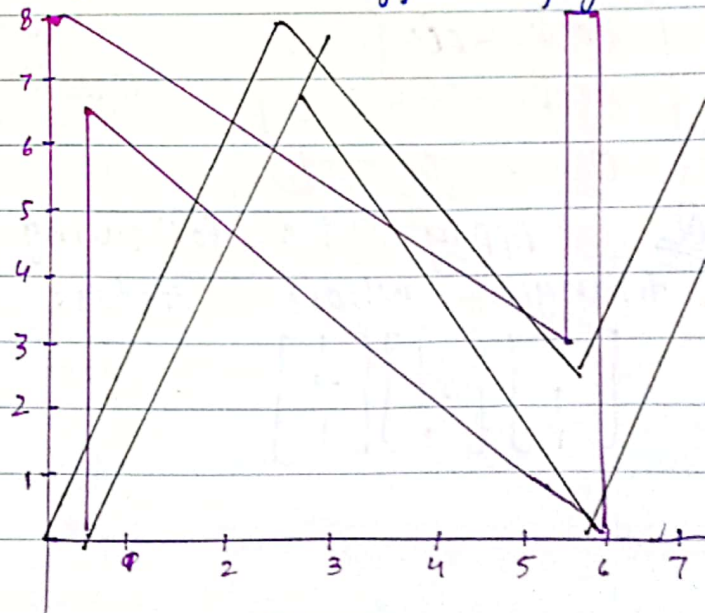
$$A_4 = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$$

$$A_5 = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$A_6 = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5.5 \\ 8 \end{bmatrix} = \begin{bmatrix} 7.5 \\ 8 \end{bmatrix}$$

$$A_7 = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5.5 \\ 1.58 \end{bmatrix} = \begin{bmatrix} 5.895 \\ 1.58 \end{bmatrix}$$

$$A_8 = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 8 \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$



Homogeneous coordinates:-

Each point (x, y) in $\mathbb{R}^2$ can be identified by a point $(x, y, 1)$ in $\mathbb{R}^3$, that lies one unit above x - y -plane. The coordinates $(x, y, 1)$ are called.....

Translation Matrix:-

"change in position is called...."

$$\begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix}$$

is a matrix that gives translation:

h = change in x -direction.

k = change in y -direction.

- Case 1:- If h is +ve obj will move in +ve x -direction.
 Case 2:- If h is -ve obj will move in -ve x -direction.
 Case 3:- If k is +ve obj will move in +ve y -direction.
 Case 4:- If k is -ve obj will move in -ve y -direction.

Scaling Matrix:-

"change in size"

$$\begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

s and s give contraction and expansion in x & y direction respectively.

s = in x -direction / s = in y -direction

Rotation Matrix:-

"Rotates matrix"

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

It rotates matrix in counter clockwise direction about z -axis.

Q. Apply the following transformation on the given triangle whose vectors are defined by the vectors.

$$\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

First:-

a Scale the given geometry by a factor 0.3. then

Ans $\begin{bmatrix} 0.3 & 0 & 0 \\ 0 & 0.3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ apply it.

b translate the given geometry by $(-0.5, 2)$ and then

$\begin{bmatrix} 1 & 0 & -0.5 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ apply it.

c rotate through an angle of 90° .

$\begin{bmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Determinant:-

Ex 1 Find the determinant of 2×2 matrix:-

$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$

$= a_{11} a_{22} - a_{12} a_{21}$

Ex 2

$|A| = \begin{vmatrix} 2 & 3 \\ 1 & 5 \end{vmatrix}$
 $= 10 - 3$

$|A| = 7$

Ex 3 $A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ Find $|A| = ?$

By expanding from the first row.

$|A| = 1 \begin{vmatrix} 4 & -1 \\ -2 & 0 \end{vmatrix} - 5 \begin{vmatrix} 2 & -1 \\ 0 & 0 \end{vmatrix} + 0 \begin{vmatrix} 2 & 4 \\ 0 & -2 \end{vmatrix}$

$= 1(0 - 2) - 5(0 - 0) + 0(-4 - 0) \Rightarrow |A| = -2$

• when one row all is zero the $|A|$ is zero • when two rows and/or two columns are zero then $|A|$ is zero.

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix} \quad |A| = -2$$

$$B = \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{bmatrix}; R_2 - 2R_1$$

$$|B| = 1 \begin{vmatrix} -6 & -1 \\ -2 & 0 \end{vmatrix} - 5 \begin{vmatrix} -1 & 0 \\ 0 & -2 \end{vmatrix} + 0 \begin{vmatrix} -6 & -2 \\ -2 & 0 \end{vmatrix}$$

$$= 1(0-2) - 5(0) + 0(0) = \boxed{|B| = -2}$$

so by performing $R_3 \pm \alpha R_1$ does not effect the value of determinant.

another:- $A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix} \Rightarrow B = \begin{bmatrix} 1 & 5 & 0 \\ 5 & 19 & -1 \\ 0 & -2 & 0 \end{bmatrix}; R_2 + 3R_1$

$$|B| = 1 \begin{vmatrix} 19 & -1 \\ -2 & 0 \end{vmatrix} - 5 \begin{vmatrix} 5 & -1 \\ 0 & 0 \end{vmatrix} + 0 \begin{vmatrix} 5 & 19 \\ 0 & -2 \end{vmatrix}$$

$$= 1(0-2) - 5(0-0) + 0 \Rightarrow \boxed{|B| = -2}$$

Ex

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 2 & 4 & -1 \end{bmatrix}; R_{23}$$

$$|B| = 1 \begin{vmatrix} -2 & 0 \\ 4 & -1 \end{vmatrix} - 5 \begin{vmatrix} 0 & 0 \\ 2 & -1 \end{vmatrix} + 0 \begin{vmatrix} 0 & -2 \\ 2 & 4 \end{vmatrix}$$

$$= 1(2-0) - 5(0-0) + 0 = \boxed{|B| = 2}$$

"If we interchange two rows than there is a change of minus sign in the value of B."

Ex

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 15 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}; 3 R_1$$

$$|B| = 3 \begin{vmatrix} 4 & -1 \\ -2 & 0 \end{vmatrix} - 15 \begin{vmatrix} 2 & -1 \\ 0 & -2 \end{vmatrix} + 0 \begin{vmatrix} 2 & 4 \\ 0 & -2 \end{vmatrix}$$

$$= 3(0-2) - 15(0-0) + 0$$

$$= -6 \Rightarrow \boxed{|B| = 3(-2)}$$

if all rows are multiplied

$$= 3 \begin{vmatrix} 12 & -3 \\ -6 & 0 \end{vmatrix} - 15 \begin{vmatrix} 6 & -1 \\ 0 & -2 \end{vmatrix} + 0 \begin{vmatrix} 6 & 12 \\ 0 & -6 \end{vmatrix}$$

$$= 3(-18) + 0 + 0 \Rightarrow -54$$

$$= 27(-2)$$

$$= 2^3(-2)$$

The $|B|$ has increased 3 times.
by 3 than ans. came as:

Ex $A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$ = The determinant of triangular matrix is just the product of its diagonal elements.

$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 5 \end{bmatrix}$ upper triangular matrix. = 10 (product)

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 1 & -1 & 3 \end{bmatrix}$ lower " " = 6

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ triangular ... = 6

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ = 0

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ = 0

$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ it is a zero matrix

$|A| = \begin{vmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{vmatrix}; R_2 - 2R_1 \Rightarrow \begin{vmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & 0 & 2/3 \end{vmatrix} R_3 - 1/3 R_2$

$\Rightarrow \begin{vmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & 0 & 1/3 \end{vmatrix} \Rightarrow (1)(-6)(1/3)$
 $\boxed{B = -2}$

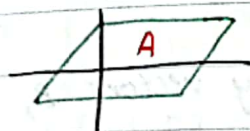
Determinant or area of a parallelogram:-

gives area of parallelogram of 2×2 matrix gives vol. of parallelepiped of 3×3 matrix

Ex 1 Find the area of parallelogram formed by vectors:
 $\begin{bmatrix} 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} 3 & 0 \\ 0 & 2 \end{vmatrix} = \boxed{6}$

Ex 2 calculate area... determined by (vectors)* four points or vertices are $[-2, -2], [0, 3], [4, -1], [6, 4]$

Add +2 to x-coordinates and +2 in y-coordinates
are $(0, 0), (2, 5), (6, 1), (8, 6)$

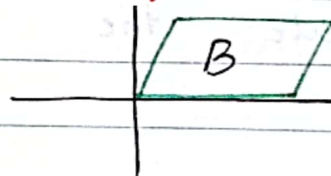


$|A| = \begin{vmatrix} 2 & 6 \\ 5 & 1 \end{vmatrix} = -28$

$|A| = \begin{vmatrix} 6 & 8 \\ 1 & 6 \end{vmatrix} = 28$

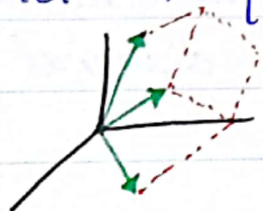
$|A| = \begin{vmatrix} 2 & 8 \\ 5 & 6 \end{vmatrix} = -28$

"so area of a parallelogram $A = ||B||$ as 28 units."



3

let $u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $u_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $u_3 = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$ Find vol. of parallelepiped formed by u_1, u_2 & u_3 .



$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 1 & 3 \end{bmatrix}$$

$$\Rightarrow |A| = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 1 & 3 \end{vmatrix} = - \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{vmatrix}$$

not $\sim$ because det. is not changed.

$$R_{23} \Rightarrow -(1)(1)(2) = -2.$$

so, vol. of parallelepiped = 2 units.

To find area of transformed figure:-

Area of transformed figure =

$\det |A| \times \text{Area of original figure.}$

Find the area of transformed figure if we have a unit circle as original geometry and consider the transformation.

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

area of original figure = π ($\pi r^2 \Rightarrow \pi(1)^2$)

$$\det |A| = 6$$

$$\text{from formula } \Rightarrow 6(\pi) = \boxed{6\pi}$$

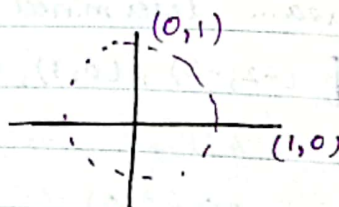
Find the transformed figure:

$$x^2 + y^2 = 1$$

$$y^2 = 1 - x^2$$

$$y = \pm \sqrt{1 - x^2}$$

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$



$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

let x be a parallelogram determined by vectors B_1 whose entries $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$, $B_2 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$. let $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$ compute the area of image of S .

$$\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1-3 \\ 0+6 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 5-1 \\ 0+2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \dots\dots$$

Inverse of matrix:-

If A and B are two matrices such that $AB = I$. Then A is inverse of B and B is inverse of A .

Ex Find the Inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 5 & 3 & 1 \end{bmatrix}$ By using row operation.

Reduced Echelon form:

If a matrix satisfies two additional conditions then the matrix is in

4th condition $\Rightarrow$ All entries above the leading entry must be zero.

5th condition $\Rightarrow$ leading entry should be 1.

Ans

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 2 & 3 & 0 & 1 & 0 \\ 5 & 3 & 1 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3/2 & 0 & 1/2 & 0 \\ 0 & -2 & -4 & -5 & 0 & 1 \end{bmatrix}; \frac{1}{2} R_2$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3/2 & 0 & 1/2 & 0 \\ 0 & 0 & -1 & -5 & 1 & 1 \end{bmatrix}; R_3 - 5R_1$$

$$\rightarrow \begin{bmatrix} 1 & 0 & -1/2 & 1 & -1/2 & 0 \\ 0 & 1 & 3/2 & 0 & 1/2 & 0 \\ 0 & 0 & -1 & -5 & 1 & 1 \end{bmatrix}; R_1 - R_2$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & 7/2 & -1 & -1/2 \\ 0 & 1 & 3/2 & 0 & 1/2 & 0 \\ 0 & 0 & -1 & -5 & 1 & 1 \end{bmatrix}; R_1 + 1/2 R_3$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & 7/2 & -1 & -1/2 \\ 0 & 1 & 0 & 15/2 & 2 & 3/2 \\ 0 & 0 & 1 & 5 & 1 & -1 \end{bmatrix}; R_2 - 3/2 R_3$$

$$A^{-1} = \begin{bmatrix} 7/2 & -1 & -1/2 \\ -15/2 & 2 & 3/2 \\ 5 & -1 & -1 \end{bmatrix}$$

جس کی مدد سے row operation کرو کرنا ہے وہ بعد میں نکھیں گے operation

Find the inverse of following matrix if exists:

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix} \Rightarrow \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix}$$

Q. If this 3×3 matrix has all rows pivot then inverse exists otherwise not.

Basis of a subspace: Basis of subspace H of $\mathbb{R}^n$ consists of all those vectors that spans H , these vectors are linearly independent.

Dimension of subspace:

The number of vectors in the set of basis is called dimension.

Rank of matrix:

The no. of non-zero rows in echelon form is called rank.

Ex.1 The vectors $\{e_1, e_2\}$ are standard basis of $\mathbb{R}^2$ where $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Ex.2 The vectors $\{e_1, e_2, e_3\}$ are standard basis of $\mathbb{R}^3$ where $e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Null space of A:-

Null space consists of all those vectors that are mapped transformed into zero vector.

Ex. Let $A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$

- Find 3- vectors in Null (A).
- Find basis of Null (A).
- Find dimension of Null (A).
- Find geometry of Null (A).

Sol (a)

$$AX=0$$

$$\begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$$

Echelon form is

$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 - 2x_2 + 2x_3 + 3x_4 - x_5 = 0$$

$$\Rightarrow x_1 + 2x_3 = 2x_2 - 3x_4 + x_5$$

$$x_3 + 2x_4 - 2x_5 = 0$$

$$\Rightarrow x_3 = -2x_4 + 2x_5$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \Rightarrow \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 + 0x_4 + 0x_5 \\ 0x_2 + 2x_4 + 2x_5 \\ 0x_2 + 1x_4 + 0x_5 \\ 0x_2 + 0x_4 + 1x_5 \end{bmatrix}$$

forms basis of Null (A).

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

($\therefore$ vectors which are present in the soln. are the basis of vector (original) that generates matrix.

c No of vectors present in the basis are dimension. So, $\dim(\text{Null } A) = 3$.

d Geometry: 3D-space in $\mathbb{R}^5$.

$$\text{Span} \{u, v, w\} = \begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} x & x & x \\ 0 & x & x \\ 0 & 0 & x \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Q What does A matrix do?

It acts on X to produce AX.



AX is parallel to X or AX is not parallel to X.

If AX is parallel to X then X is called Eigen vector

$$AX = \lambda X$$

Eigen vector Eigen value.

$$\text{let } A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}, u = \begin{bmatrix} 6 \\ -5 \end{bmatrix}, v = \begin{bmatrix} 3 \\ -2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \times 6 - 30 \\ 5 \times 6 - 10 \end{bmatrix}$$

$$A = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4 \begin{bmatrix} 6 \\ -5 \end{bmatrix} \text{ so, } Au = -4u$$

so, $\begin{bmatrix} 6 \\ -5 \end{bmatrix}$ is the Eigen vector & $\lambda = -4$ is eigen value.

$$A = \begin{bmatrix} 3 & -12 \\ 15 & -4 \end{bmatrix} \Rightarrow \begin{bmatrix} -9 \\ 11 \end{bmatrix} \Rightarrow AV \neq \lambda V.$$

$$AX = \lambda X$$

$$AX - \lambda X = 0$$

$$(A - \lambda I)X = 0 \Rightarrow \text{To find Eigen vector } X.$$

$$\text{To find } \lambda \Rightarrow |A - \lambda I| = 0 \rightarrow \text{To find eigen value.}$$

Ex Find Eigen vector & value of $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$

$$|A - \lambda I| = 0$$

$$\left| \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 6 \\ 5 & 2-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(2-\lambda) - 30 = 0$$

$$\lambda^2 - 3\lambda - 28 = 0$$

$$\lambda = \frac{3 \pm \sqrt{9 - 4(1)(-28)}}{2}$$

$$\text{so, } \lambda = 7, -4.$$

$$\therefore \lambda = 7$$

$$\left(\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) X = 0 \Rightarrow \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix} X = 0$$

$$= \begin{bmatrix} -6 & 6 \\ 0 & 0 \end{bmatrix} \Rightarrow -6x_1 + 6x_2 = 0$$

$$x_1 = x_2$$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ or } X = x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ or } \begin{bmatrix} 2 \\ 2 \end{bmatrix} \text{ or } \begin{bmatrix} 3 \\ 3 \end{bmatrix} \dots$$

$$\therefore \lambda = -4 \Rightarrow \left(\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - (-4) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) X = 0 \Rightarrow \begin{bmatrix} 5 & 6 \\ 5 & 6 \end{bmatrix} X = 0 \Rightarrow \begin{bmatrix} 5 & 6 \\ 0 & 0 \end{bmatrix} X = 0$$

$$5x_1 + 6x_2 = 0 \Rightarrow x_1 = -\frac{6}{5}x_2$$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{6}{5}x_2 \\ x_2 \end{bmatrix} \therefore \text{if } x_2 = 1 \Rightarrow X = \begin{bmatrix} -6/5 \\ 1 \end{bmatrix}$$

Eigen space:-

The collection of all eigen vectors is called an eigen space.

c what are the basis for Eigen space? formed by considering $\lambda = 7$?

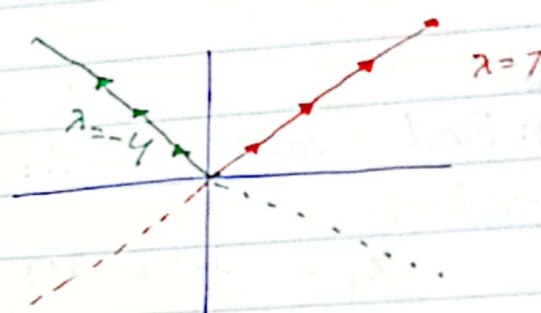
$$\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

This vector forms an eigen space corresponding to $\lambda = 7$.
 $\dim(\text{Eigen space}) = 1$

d what is the geometry of it? Line in $\mathbb{R}^2$.

All is same by $\lambda = -4$ but they have different directions of lines due to -ve sign.

• if this transformation is applied on a than vector placed on the lines than the magnitude of vectors is increased by 7 or -4 times (by λ).



1c1 Find Eigen values of given matrix A.

a ... Eigen vectors corresponding to each eigen value.

b Find eigen spaces, basis of eigen spaces & dimensions.

$$A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$$

$$\left| \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right| = 0 \Rightarrow \begin{vmatrix} 4-\lambda & -1 & 6 \\ 2 & 1-\lambda & 6 \\ 2 & -1 & 8-\lambda \end{vmatrix} = 0$$

$$(4-\lambda) \begin{vmatrix} 1-\lambda & 6 \\ 2 & 8-\lambda \end{vmatrix} - (-1) \begin{vmatrix} 2 & 6 \\ 2 & 8-\lambda \end{vmatrix} + 6 \begin{vmatrix} 2 & 1-\lambda \\ 2 & -1 \end{vmatrix} = 0$$

$$(4-\lambda) [(1-\lambda)(8-\lambda)+6] + [2(8-\lambda)-12] + 6[-2-2(1-\lambda)] = 0$$

$$(4-\lambda) [\lambda^2 - 9\lambda + 14] + [-2\lambda + 4] + 6[2\lambda - 4] = 0$$

$$4\lambda^2 - 36\lambda + 56 - \lambda^3 + 9\lambda^2 - 14\lambda - 2\lambda + 4 + 12\lambda - 24 = 0$$

$$-\lambda^3 + 13\lambda^2 - 40\lambda + 36 = 0$$

$$\lambda^3 - 13\lambda^2 + 40\lambda - 36 = 0$$

$$\begin{array}{c|cccc} & 1 & -13 & 40 & -36 \\ 2 & 0 & 2 & -22 & 36 \\ \hline & 1 & -11 & 18 & 0 \end{array}$$

add then multiply.

$\lambda = 2$ is a root of eq.

$(\lambda - 2)(\lambda^2 - 11\lambda + 18) = 0$ use quadratic formula to find another two values.
So, $\lambda = 2, \lambda = 2, \lambda = 9$.

- Here we have 3 eigen values. $\lambda = 2$ is an eigen value of multiplicity 2 (two). $\lambda = 9$ is an eigen value of multiplicity 1.
- The eq. with root in decimal like 1.1, 5.0... cannot be calculated through standard deviation.

Q: Find value of X :

$\therefore$ for $\lambda = 2$

$$x_1 = \frac{1}{2}x_2 - 3x_3$$

$$x_2 = x_2, \quad x_3 = x_3$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \dots \\ \dots \\ \dots \end{bmatrix}$$

$$X = x_2 \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

(ii)

$$\lambda = 9 \quad X = x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$x_3 = 1$$

$$x_2 = \frac{9}{7} \quad 1$$

$$x_1 = \frac{37}{35} \quad 1$$

(c) Find ^{basis} eigen of eigen spaces corresponding $\lambda = 2$.

$$\left\{ \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Find dimensions to $\lambda = 2 \Rightarrow$

Find geometry of $\lambda = 2 \Rightarrow$ plane in $\mathbb{R}^3$. $\dim(\text{Eigen space}) = 2$.

Find basis of Eigen space of $\lambda = 9 \Rightarrow$

Find dimension to $\lambda = 9 \Rightarrow \dim(\text{Eigen space}) = 1$

Find geometry of $\lambda = 9 \Rightarrow$ line in $\mathbb{R}^3$.

Similar Matrices:

Two matrices A & D are similar if their properties are same like determinant, eigen values, trace, characteristic eq. and so on.

Mathematical definition of similar matrices:-

Two matrices A & D are similar if there exists an invertible matrix P such that $A = PDP^{-1}$.

If A & D are similar then prove that $\det(A) = \det(D)$.

Sol A & D are similar.

$$\Rightarrow A = PDP^{-1}$$

$$\Rightarrow \det(A) = \det(PDP^{-1}) \Rightarrow \det(A) = \det(P) \cdot \det(D) \cdot \det(P^{-1})$$

$$\Rightarrow \det(A) = \det(D) \cdot \det(P) \cdot \det(P^{-1}) \Rightarrow \det(A) = \det(D) \cdot \det(PP^{-1}) \Rightarrow \det(A) = \det(D) \cdot \det(I)$$

$$\boxed{\det(A) = \det(D) \cdot 1}$$

Q How to find a similar matrix D of A .

Def:-

The process of computing similar matrix D of A is called diagonalization.

Ex

Diagonalize A where $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ $D = ?$

Means you have to compute a diagonal matrix D of A where both A & D are similar.

Step I Eigen values are $\lambda = -2, \lambda = -2, \lambda = 1$

Step II Eigen vector's $= x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ for $\lambda = -2$, $x_1 = x_3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ For $\lambda = 1$

Step III write D from eigen value.

$$D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

when $\det = 0$ then the matrix/vectors are linearly dependent. opposite.

if $x_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$
 $x_3 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$

$P = \begin{bmatrix} -1 & ? & 1 \\ 1 & ? & -1 \\ 0 & ? & 1 \end{bmatrix}$



It becomes linearly dependent so inverse of P doesn't exist.

Q IV

for P all row are pivot and has unique soln.

write P from eigen vector:-

$$P = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$A = PDP^{-1} \Rightarrow AP = PD P^{-1}P$$

$$AP = PD$$

$$A = PDP^{-1}$$

$$A^2 = (PDP^{-1})(PDP^{-1})$$

$$= PD(P P^{-1}) D P^{-1}$$

$$= P D D P^{-1} \Rightarrow \boxed{A^2 = P D^2 P^{-1}}$$

As we know that.

$$A^k = P D^k P^{-1}$$

$$A^{500} = P D^{500} P^{-1}$$

$$P = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$D^{500} = \begin{bmatrix} 2^{500} & 0 & 0 \\ 0 & 2^{500} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{500} = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2^{500} & 0 & 0 \\ 0 & 2^{500} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} P^{-1} \end{bmatrix}$$

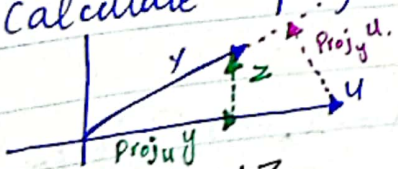
Q Prove that A & D are similar A is given & D is given.

Q Is A diagonalizable? If yes find the diagonal matrix D.

Q Find A^{500} ? Or Find A^{700} of given A.

Orthogonal Projection:-

Calculate projection of y onto u.



$$y = \alpha u + z$$

$$z = y - \alpha u$$

z is orthogonal to u.

$$\text{so, } z \cdot u = 0$$

$$(y - \alpha u) \cdot u = 0$$

$$\text{proj. of } y \text{ onto } u = \text{Proj}_u y = \alpha u \Rightarrow \frac{y \cdot u}{u \cdot u} u$$

$$y \cdot u - \alpha \cdot u \cdot u = 0$$

$$\alpha u \cdot u = y \cdot u$$

$$\alpha = \frac{y \cdot u}{u \cdot u}$$

$$\boxed{\alpha = \frac{y \cdot u}{u \cdot u}}$$

Basis, mathematical criteria to prove that given set forms bases, properties of basis vectors.
 independent span is always equals to geometry.



Two vectors are orthogonal to each other if $u \cdot v = 0$.

Ex: 1

if $u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

$$u \cdot v = u^T \cdot v = \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \Rightarrow u_1 v_1 + u_2 v_2 + u_3 v_3.$$

Proj_y $u = \frac{u \cdot y}{y \cdot y} y$.

Let $y = \begin{bmatrix} 7 \\ 6 \end{bmatrix}, u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$

$$y^T \cdot u = \begin{bmatrix} 7 & 6 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 40$$

$$u^T \cdot u = \begin{bmatrix} 4 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 20$$

$$\Rightarrow \frac{40}{20} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}$$

Orthogonal sets:-

"A set $\{u_1, u_2, u_3\}$ is said to be orthogonal if every vector is orthogonal to remaining vectors."

Orthonormal set:-

"A set $\{u_1, u_2, u_3\}$ is said to be orthonormal if every vector is (perpendi) orthogonal to remaining vectors and every vector should be a unit vector."

* if $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$, find $A^{100} = ?$

sol: Step I:- Find eigen values = ?

$$\lambda = -2, \lambda = -2, \lambda = 1.$$

Step III:-

$$D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, x$$

Step II:- Find eigen vectors = ?

$$\lambda = -2; \quad x = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda = 1; \quad x = x_3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Ex Prove that set $\{v_1, v_2, v_3\}$ forms an orthonormal set.
Where $v_1 = \begin{bmatrix} \frac{3}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \end{bmatrix}$, $v_2 = \begin{bmatrix} \frac{-1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$, $v_3 = \begin{bmatrix} \frac{-1}{\sqrt{66}} \\ \frac{-4}{\sqrt{66}} \\ \frac{7}{\sqrt{66}} \end{bmatrix}$

sol $v_1 \cdot v_2 = 0$, $v_2 \cdot v_3 = 0$, $v_1 \cdot v_3 = 0$

$$||v_1|| = \sqrt{\frac{9}{11} + \frac{1}{11} + \frac{1}{11}} = 1 \Rightarrow \sqrt{\frac{x_1^2}{\dots} + x_2^2 + x_3^2}$$

$$||v_2|| = \sqrt{\frac{1}{6} + \frac{4}{6} + \frac{1}{6}} = 1$$

$$||v_3|| = \sqrt{\frac{1}{66} + \frac{16}{66} + \frac{49}{66}} = 1$$

Gram-Schmidt Process:-

Let $\{u_1, u_2, \dots, u_p\}$ are simple basis for any subspace w of $\mathbb{R}^n$. Then we can find its orthogonal and orthonormal basis by this process."

Let $\{u_1, u_2, \dots, u_p\}$ are simple basis of w .

Let $\{v_1, v_2, \dots, v_p\}$ are orthogonal basis of w .

Let $\{\hat{v}_1, \hat{v}_2, \dots, \hat{v}_p\}$ are orthonormal basis of w .

} all of w has same span.

Formulas are:-

$$v_1 = u_1$$

$$v_2 = u_2 - \frac{u_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$v_3 = u_3 - \frac{u_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{u_3 \cdot v_2}{v_2 \cdot v_2} v_2$$

Ex If $\{u_1, u_2, u_3\}$ $u_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$, $u_3 = \begin{bmatrix} 1 \\ 5 \\ -1 \end{bmatrix}$

$$v_1 = u_1$$

$$v_2 = \begin{bmatrix} -17/14 \\ 11/7 \\ -9/14 \end{bmatrix}$$

$v_3 =$ Find

Ex. 2

$$W = \text{span} \{u_1, u_2\}$$

where $u_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

W is plane in $\mathbb{R}^3$, generated by u_1 & u_2 . calculate (a) & (b).

a orthonormal:-

$$v_1 = u_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$$v_2 = u_2 - \frac{u_2 \cdot v_1}{v_1 \cdot v_1} v_1 \Rightarrow$$

$$u_2 \cdot v_1 = u_2^T v_1 = [1 \ 2 \ 2] \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = 15$$

$$v_1 \cdot v_1 = v_1^T v_1 = [3 \ 6 \ 0] \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = 45$$

$$v_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \frac{15}{45} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

b Orthonormal basis?

$$\hat{v}_1 = \frac{v_1}{|v_1|}, \quad \hat{v}_2 = \frac{v_2}{|v_2|}$$

$$\hat{v}_1 = \frac{1}{\sqrt{45}} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 3/\sqrt{45} \\ 6/\sqrt{45} \\ 0 \end{bmatrix}$$

$$\hat{v}_2 = \frac{1}{2} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

c Prove that $\{u_1, u_2\} = \text{span}\{v_1, v_2\} = \text{span}\{\hat{v}_1, \hat{v}_2\}$

Ans $\text{span}\{u_1, u_2\} = \text{plane in } \mathbb{R}^3$.

$$\text{span}\{v_1, v_2\} = "$$

$$\text{span}\{\hat{v}_1, \hat{v}_2\} = "$$

To prove that both planes are same. We have to prove

these six vectors $u_1, u_2, v_1, v_2, \hat{v}_1, \hat{v}_2$ are coplanar.

$$\begin{bmatrix} | & | & | & | & | & | \\ u_1 & u_2 & v_1 & v_2 & \hat{v}_1 & \hat{v}_2 \\ | & | & | & | & | & | \end{bmatrix} \text{ there should be two pivots so they have plane in } \mathbb{R}^3.$$

Ex. 3 $u_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$ $u_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$ $y = \begin{bmatrix} 3 \\ 6 \\ 1 \end{bmatrix}$

$\text{span} \{ u_1, u_2 \} = W =$ plane in $\mathbb{R}^3$.

Find $\text{proj}_W y = ?$

$$\text{proj}_W y = \text{proj}_{u_1} y + \text{proj}_{u_2} y$$

$$= \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{y \cdot u_2}{u_2 \cdot u_2} u_2 = \begin{bmatrix} \\ \\ \end{bmatrix}$$

$$y = \text{proj}_W y + z$$

$$z = y - \text{proj}_W y$$

$$z = y - \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 - \frac{y \cdot u_2}{u_2 \cdot u_2} u_2$$